

The control panel shall receive 230 V A/C supply & interconnecting wiring between control panel & actuator shall be done using fireproof cables.

Access door will be provided in the duct before each fire damper.

13.9 DUCTING INSULATION

Insulation Procedure in case of Cross Linked Polyethylene Duct Insulation the Instruction to be followed are:

- The Insulation material for Ducts shall be closed cell Cross-Linked Polyethylene Foam.
- The Thermal Conductivity of the material shall not exceed 0.036 W/Mk at an average temperature of 40°C.
- The product will have bending Trial and the Dimensional Stability as per DIN 51949 and DIN 53431 for an operating range of - 40°C to + 110°C.
- The Density of the Material shall be 30 ± 3 Kg/m³ or 0.30 GM/CC.
- The SMOKE DENSITY of the Material as per AS-1530.3 shall not exceed 1.
- The Water Vapour Permeability as per DIN 52615 shall not exceed 0.15 ng/m.sec.Pa
- The material shall have a Fire Approval from CBRI/Fire Advisor (Govt. of India) / Chief Fire Officer.
- For providing UV protection the insulation shall be cladded with minimum 30 micron Aluminum P.E. foil. The cladding shall be factory finished to avoid site work. The minimum thickness shall be as per specifications. The insulation material will be provided with self adhesive covered peel off paper (as an alternative).

Insulation Procedure of Nitrile Rubber Duct Insulation the Instruction to be followed are:

- The Insulation material for Ducts shall be Elastomeric Nitrile rubber .
- The Thermal Conductivity of the material shall not exceed 0.035 W/Mk at an average temperature of 40°C.
- The product will have bending Trial and the Dimensional Stability as per DIN 51949 and DIN 53431 for an operating range of 0°C to + 105°C.
- The Density of the Material shall be 30 ± 3 Kg/m³ or 0.30 GM/CC.
- The SMOKE DENSITY of the Material as per AS-1530.3 shall not exceed 1.
- The Water Vapour Permeability as per DIN 52615 shall not exceed 0.15 ng/m.sec.Pa
- The material shall be fire rated for Class 0 as per BS 476 Part 6 : 1989 or FM global approved for fire propagation test and for Class 1 as per BS 476 Part 7, 1997 for surface spread of flame test
- For providing UV protection the insulation shall be cladded with minimum 30 micron Aluminum P.E. foil. The cladding shall be factory finished to avoid site work. The minimum thickness shall be as per specifications. The insulation material will be provided with self adhesive covered peel off paper (as an alternative).

Comply

Comply with
Deviation
Density will
40-60 kg/m³
K value 0.038
w/mk at 40 C
means
temperature

Application:

The Duct surfaces will be thoroughly cleaned prior to applying the insulation. The insulation shall be provided with self-adhesive; otherwise the Adhesive of suitable grade shall be uniformly applied on the insulation and cured before, sticking to the duct. The insulation can be wrapped around the duct as one piece, where size does not permit the same be cut to exact width /height of duct. Where size does not permit the same to be cut to exact Width / Height of duct.

The duct insulation joint will be overlapped with a self-adhesive tape of the same material. The tape shall be minimum 2.5mm thick and 50mm wide.

A self-adhesive strip of same material of suitable thickness, to over the complete height of the flange will be provided around the flanges and the joint neatly covered with insulation, so as to reduce heat loss through flanges, in addition to covering the flange connection/joint.

13.10 PIPING INSULATION**13.10.1 Chilled Water & Hot Water Line Insulation****Material**

Sr. No.	Item	Description
1	Services	Chilled water/ Hot Water
2	Material	FM approved Elastomeric Nitrile Rubber, closed cell, Class 'O' fire Propagation as per BS 476 Part 6 and Surface spread of flame Class 1 as Per BS 476 Part 7.
3	Density	40 kg/m ³
4	Thermal Conductivity (K)	Shall not exceed 0.035 w/m°K at Average temp of 0°C
5	Chemically Treated Woven Glass Cloth	7 mil thickness, Density: 205 + 10 kg / m ³ .
6	Adhesive	Recommended by Insulation material manufacturer.

Inspection

Following physical checks and verifications shall be done on site by Consultant/ Client's representative on site

- Check Scrutiny of Test Certificates with specification requirements.
- Check Raw material identification against material Test Certificate from manufacturer.
- Visual and final dimensional inspections as per relevant standards.

Application of insulation

- Insulation shall be applied after all Hydro test/leak test on piping & equipment are over and the portion of the plant has been specifically released by the Client/Consultant for such work. If insulation has to be applied before the Hydro test/leak test, all welded and

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flanged joints shall be left exposed and insulated after satisfactory completion of Hydro test/leak test.

- Before application of the insulation, the surface shall be wire brushed to remove all dirt, rust, scale oil etc. and dried.
- Particular attention should be made during cleaning in Service shafts where concrete often sits on the pipework and pipe-supports during building construction.
- On piping, slit opened tube of the insulating material (slit with a very sharp Knife in a Straight line) shall be placed over the pipe and adhesive shall be applied.
- Adhesive must be allowed to tack dry and then press surface firmly together starting from butt ends and working towards center.
- Wherever flat sheets shall be used it shall be cut out in correct dimension. All longitudinal and transverse joints shall be sealed as per manufacturer recommendations. The insulation shall be continuous over the entire run of piping, fittings and valves.
- Apply the Nitrile Rubber in the 2 m lengths where possible. Fix the Nitrile Rubber on to the dry pipe and apply adhesive to the fixing seams, ensure all surface seams are covered with the adhesive.
- All the Nitrile Rubber should be installed under compression. All tube should have an extra 5-10 mm added to all cut lengths to allow for this compression.
- All compressed butt joints should be wet sealed with the Nitrile Rubber adhesive at all times. Self-adhesive black electrical tape should never be used on Nitrile Rubber to provide a vapour seal.
- All Nitrile Rubber tube/sheet ends should be glued to the pipework at termination points in locations of valves and fittings and vertical pipework installations.
- All pipe-supports shall be vapour sealed with Nitrile Rubber adhesive, ensure no gaps are present at such locations.
- All valves, fittings, joints, strainers, etc. in piping shall be insulated to the same thickness as specified for the main run of piping and application shall be same as above. Valves bonnet, yokes and spindles shall be insulated in such a manner as not to cause damage to insulation when the valve is used or serviced. If there are joints that are hard to get too, the use of a good quality Silicone Sealant could be used to provide the vapour seal. Ensure all exposed metal of the fittings is insulated to prevent condensation
- Unless specified otherwise in the tender Valves & Strainers of sizes 100 mm NB and larger shall be provided with removable box type insulation. Flanged joints shall also be insulated with removable type of boxes. A typical arrangement is as per sketch attached.
- All equipment manholes, cover plate, flanged ends etc. shall have removable box type insulation with same thickness of insulation as for adjacent surfaces.
- Name plate on equipment shall not be insulated. Proper care shall be taken to eliminate seepage of moisture from such un-insulated portion into the insulating material.

- Avoid keeping the adhesive brushes in the adhesive holder during application. Always keep a cover on the adhesive when not using. Keep adhesive brushes stored clean when not in use.
- Avoid damaging the Nitrile Rubber during application and fabrication. Keep the Nitrile Rubber clean during fabrication where appropriate fabrication and adhesive work should be done off the ground.
- After application of Nitrile Rubber clean the dust from all visual surfaces.
- Where possible keep the Nitrile Rubber seams away from view and facing towards the ceiling and not the floor, for better look.

Application of insulation for Valves and Flanges:

- On these fittings always maintain the same insulation thickness as the main pipework branches.
- Always-wet seal all joints with Armaflex adhesive. If there are joints that are hard to get too, the use of a good quality Silicone Sealant could be used to provide the vapour seal.

Ensure all exposed metal of the fittings is insulated to prevent condensation. In area and locations that have exposed electrical equipment, Valve Handle shall be insulated with removable Armaflex fabrications.

Application procedure for Chemically Treated Woven Glass Cloth

- Glass Cloth should be cut to size according the circumference of the insulated pipe. Additional 50 mm for the overlap of the covering material.
- The Overlap shall be securely fixed using synthetic contact adhesive.
- Stick the seam of the covering to the insulation with the help of adhesive and then, spread thin film of adhesive on the 50 mm overlap and close the seam.
- All fittings have to be covered to the same standards as the covering of the pipe work. No additional vapor barrier is needed.
- Application of Glass cloth should be carried out after 24 hrs. of application of Nitrile Rubber but not later than 72 hrs.
- For outdoor application mastic must be applied on overlap seams, in addition to the adhesive.

13.10.2 Hot Insulation (If Applicable)

This specification covers the technical requirement and essential particulars for the supply, application and finishing of complete thermal insulation for hot service piping and equipment. The scope of supply of the contractor shall include, but not be limited to the following items;

- Insulation material of all types as specified and required
- Insulation adhesives, vapour barriers and finishing material of all types as specified and required

- Auxiliary material such as binding and lacing wires, wire netting, bands, screws etc. as specified and required
- Any other material as may be required for making the insulation complete

Material

- Service : High pressure & Low pressure steam line/Steam condensate/ Hot water
- Material : Lightly resin bonded mattress of minerals wool as per IS 8183 (Part 3)
- Chloride Content : Less than 35 PPM
- Density : 144 Kg/ M3

Cladding

Aluminium sheet as per IS 737, 22 SWG, 99.5% Pure Aluminum sheets.

Band

The cladding shall be fixed with Aluminum band of size 20 mm x 20 SWG at a distance of 450 mm center to center.

Wire netting

20 mm Mesh 20 SWG galvanized wire netting in tortoise pattern.

Lacking wire

20 SWG Galvanized soft iron wire.

Aluminum foil

0.1 mm thick Aluminum foil shall be wrapped on all SS vessels/ pipe before the application of Hot Insulation.

Screws

Self-tapped screws shall be used on Hot Insulation cladding.

Inspection

Following physical checks and verifications shall be done on site by Consultant/ Client's representative on site

- Check Scrutiny of Test Certificates with specification requirements.
- Check Raw material identification against material Test Certificate from manufacturer.
- Visual and final dimensional inspections as per relevant standards.

Application of hot insulation

- Clean the surface to be insulated to remove dirt, dust, grease etc.

- Check the equipment surface where MS flat ring with spike fixed or not for Clamping of mattress by wire. (for Flat surface insulations only)
- LRB Mattress having density of 144 kg/ m³ with one side 20 mm x 24 SWG GI wire netting in tortoise pattern & stitch with 18 SWG Galvanized soft iron wire.
- Apply cotton canvas with salikot as vapours barrier (For Temperature below 125 °C).
- Fabricate & Fix 22 SWG of Aluminum sheet with GI self-tapping screw at 150 mm distance as the case may be with proper grooves & adequate overlapping (minimum 50 mm).
- While applying flexible materials care shall be taken to ensure that the material is applied at the recommended density.
- While applying multi-layer insulation all joints shall be staggered; and each layer shall be separately secured to the surface.
- In consideration of possible pipeline movement with change in fluid Temperatures, different pipes should be insulated separately.
- The insulation shall be supported when applied to the sides of underneath large vessels or to long runs of vertical piping. Supports shall be cleats, studs, washers, nuts, bolts, lugs, pins or collars (rings) which shall be either welded to the hot surface or to bands which are then strapped round the surface. These supports serve to hold the insulation in place, prevent its slipping or support it above expansion joints. In addition, they shall provide necessary anchorage for lacing wire or wire netting which may be required to hold the insulation in place and / or to provide reinforcement for the insulation or a finishing material. Depending on their function, supports shall either penetrate only partly through the insulation or protrude slightly beyond it. But in no case the supports shall protrude through the finish.
- As a rule fitting on vessels shall be covered with an independent insulation so as to allow easy access and removal without disturbing the main insulation.
- Stiffener angles, weld protrusions ladder supports, insulation support rings, pipe hangers or any metal connections not otherwise scheduled to receive insulation shall be insulated if in direct contact with the hot surface. Thickness of insulation on such protrusions shall be not less than 50% of the thickness of the main system.
- All austenitic stainless steel surfaces, proposed to be insulated and subjected to hot insulation shall be suitably protected, such as by wrapping with aluminum foil, 0.1 mm thick before application of insulation.

Application of Insulation over Valves and Flanges:

- On these fittings always maintain the same insulation thickness as the main pipework branches.
- Always-wet seal all joints with Armaflex adhesive. If there are joints that are hard to get too, the use of a good quality Silicone Sealant could be used to provide the vapour seal.

- Ensure all exposed metal of the fittings is insulated to prevent condensation. In area and locations that have exposed electrical equipment, Valve Handle shall be insulated with removable Armaflex fabrications.

13.10.3 Measurement of Insulation

The Schedule of Quantities and Unit Rates unless otherwise stated shall be deemed to have been prepared in accordance with the Standard Method of Measurement as per Indian Standard IS-14164.

The payment for supply & installation will be made as per actual installed & measured quantity. No any extra payment shall be made for reject or excess material at site.

All measurements for piping shall be taken over the finished insulated surface in meters, corrected to nearest centimeter along the center line of piping, through all fittings, insulated or otherwise, such as valves, flanges, elbows, bends, tees and reducers.

Insulation of valves, flanges or other fittings shall be measured through such valves, Flanges or other fittings as per above and in addition extra measurements shall be allowed as follows:

- For each insulated valve including flanges and body 1.50 linear meters if piping of the connected line size up to 300mm line size and 2.00 linear meters for larger sizes.
- For each pair or insulated flanges including orifice plate and flanges, 0.80 linear meters of piping of the connected line size.
- All insulated bends, elbows and tees twice the actual length of such fittings, of concerned connected line sizes, as measured along their center lines. In case of insulated welded, tees twice the diameter over insulation of each of the two pipes forming the tee is to be added to the overall length.
- Any other special fittings not covered under the above shall be specified and measured separately on number basis.
- Insulated reducers connecting two different diameters of pipes shall be reckoned as equivalent piping of larger diameter for purposes of measurement of insulation.
- Anti- corrosive painting or wrapping with aluminum foil over stainless steel / alloy steel piping and equipment prior to application of insulation shall be included in insulation unit rate cost.
- Insulation of instruments shall be measured separately and the length covered by such instruments which are not axial with the pipeline shall not be considered in the measurement of the connected piping insulation.
- Inspection plugs, if any provided in insulation of piping or any equipment shall be measured separate on number basis and no deduction shall be made on this account from the overall – measurement of insulation of the concerned equipment or piping.

Valuation of completed work shall be based on the schedule of rates applied to the assessed work content of general arrangement and isometric drawings and to the actual work content of field run pipe work and includes.